NIFTY50 Quant Research Report

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1. Project Overview

This project implements an institutional-style quantitative research framework for the NIFTY50 index. It performs multi-strategy backtesting on India's top stocks using four trading strategies: EMA Crossover, RSI, MACD, and Bollinger Bands. The framework includes benchmark comparison, risk analysis, and portfolio analytics.

2. Data Overview

Benchmark: ^NSEI (NIFTY50)
Date Range: 2007-09-17 to 2026-05-22 (Common Start Date)
Stocks Analyzed: 3 (RELIANCE.NS, TCS.NS, INFY.NS)
Data Source: Yahoo Finance (yfinance)

3. Strategy Descriptions

EMA Crossover:

Buys when fast EMA(20) crosses above slow EMA(50), sells on cross below.

RSI S

Buys when RSI(14) < 30 (oversold), sells when RSI > 70 (overbought).

MAC

Buys when MACD (12-26 EMA diff) > 0, sells when MACD < 0.

Bolli

Buys when price < lower band (SMA-2*std), sells when price > upper band (SMA+2*std).

4. Backtest Results Summary

Stock	Strategy	Equity($)	Return%	CAGR%	Sharpe	MaxDD%	WinRate%	Profit
RELIANCE	EMA	$72,221	-27.78%	-1.19%	-0.0841	-80.69%	25.0%	1.0397
RELIANCE	RSI	$391,901	291.9%	5.17%	0.3031	-57.11%	72.88%	2.2436
RELIANCE	MACD	$141,278	41.28%	1.28%	0.0873	-66.8%	32.14%	1.3764
RELIANCE	BOLLINGER	$203,972	103.97%	2.67%	0.161	-59.76%	64.15%	1.7454
TCS	EMA	$310,124	210.12%	4.27%	0.2801	-47.23%	31.25%	2.2794
TCS	RSI	$305,299	205.3%	4.21%	0.2941	-49.71%	82.76%	2.7116
TCS	MACD	$325,878	225.88%	4.46%	0.2977	-51.58%	34.12%	2.0559
TCS	BOLLINGER	$317,579	217.58%	4.36%	0.297	-50.52%	78.85%	1.9045
INFY	EMA	$221,020	121.02%	2.97%	0.1934	-52.17%	30.0%	2.0092
INFY	RSI	$149,303	49.3%	1.49%	0.1049	-57.89%	62.96%	1.4834
INFY	MACD	$428,858	328.86%	5.53%	0.3715	-36.7%	42.86%	2.2338
INFY	BOLLINGER	$87,743	-12.26%	-0.48%	-0.0342	-50.02%	63.27%	1.1417

5. Performance Analysis

Best Overall Return: INFY with MACD strategy (328.86% return, CAGR 5.53%)
Best CAGR: INFY with MACD (5.53%)
Best Risk-Adjusted (Sharpe): INFY with MACD (Sharpe: 0.3715)
Worst CAGR: RELIANCE with EMA (-1.19%)

Strategy Averages Across Stocks:

Strategy	Avg Return%	Avg CAGR%	Avg Sharpe	Avg MaxDD%	Avg WinRate%
EMA	101.12%	2.02%	0.1298	-60.03%	28.75%
RSI	182.17%	3.62%	0.2340	-54.90%	72.87%
MACD	198.67%	3.76%	0.2522	-51.69%	36.37%
BOLLINGER	103.10%	2.18%	0.1413	-53.43%	68.76%

6. Conclusions

1. MACD Strategy leads across all metrics with the highest average CAGR (3.76%) and best risk-adjusted returns (Sharpe 0.2522), making it the most consistent strategy.

2. RSI Strategy excels on RELIANCE (5.17% CAGR) but struggles on INFY (only 1.49% CAGR), indicating stock-specific suitability - works best on mega-cap liquid stocks.

3. Bollinger Bands shows mixed results - strongest on TCS (4.36% CAGR) but negative on INFY (-0.48% CAGR), highlighting the need for stock selection.

4. EMA Crossover delivers moderate returns on TCS/INFY but negative on RELIANCE (-1.19% CAGR) with higher drawdowns, typical of trend-following in range-bound markets.

5. No single strategy consistently outperforms across all stocks - strategy selection should be stock-specific and market-regime aware.

6. Buy & Hold returns generally exceed strategy returns over the full 18-year period, highlighting the challenge of market timing vs. long-term holding in a bull market.

7. Recommendations

1. Use MACD Strategy as a baseline for multi-stock portfolios - it offers the best risk-adjusted returns across different stocks.

2. Apply RSI Strategy selectively - excellent for large-cap liquid stocks (RELIANCE, TCS) but avoid on more volatile growth stocks (INFY).

3. Consider parameter optimization - the default parameters (20/50 EMA, 14 RSI, etc.) may not be optimal. Grid search over parameter space could improve returns.

4. Combine strategies into an ensemble model - for example, use MACD for trend direction confirmation and RSI for entry timing.

5. Add risk management rules - position sizing, trailing stop-losses, and portfolio-level risk limits to reduce drawdown severity.

6. Extend analysis to all 15 NIFTY50 stocks in the config and include sector-wise performance decomposition.

7. Incorporate machine learning models (e.g., Random Forest, XGBoost) for regime detection to dynamically select the best strategy.

8. Backtest with transaction costs and slippage models more realistic than the current 0.2% flat commission.